



COMPLETE STUDY NOTES

Data Communications and Networking

Units 1 & 2 — Exam Preparation Guide



Chapters 1, 2, 3, 6 & 7

Theory, Visual Diagrams, Worked Examples & Practice Questions

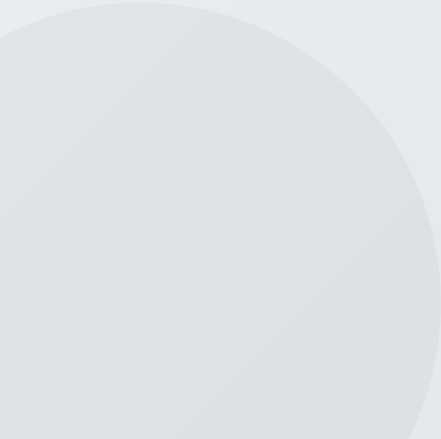


Table of Contents

Chapter 1: Data Communications

- 1.1 Fundamentals of Data Communications
- 1.2 Networks and Network Criteria
- 1.3 Network Types and Switching

Chapter 2: Network Models

- 2.1 Protocol Layering
- 2.2 TCP/IP Protocol Suite
- 2.3 The OSI Model

Chapter 3: Data and Signals

- 3.1 Analog and Digital Signals
- 3.2 Periodic Analog Signals
- 3.3 Digital Signals
- 3.4 Transmission Impairment
- 3.5 Data Rate Limits

Chapter 6: Bandwidth Utilization

- 6.1 Multiplexing Techniques

Chapter 7: Transmission Media

- 7.1 Introduction to Transmission Media
- 7.2 Guided Media
- 7.3 Unguided Media

Practice Questions & Answers

- Theoretical Questions
- Numerical Problems

Chapter 1: Data Communications

Data communications forms the foundation of modern networking. It refers to the exchange of data between two devices via some form of transmission medium, such as a wire cable. For communication to occur, the devices must participate within a communication system made up of hardware (physical equipment) and software (programs). The effectiveness of any data communications system depends on four fundamental characteristics that govern how well information flows from source to destination.

Four Fundamental Characteristics

Delivery. The system must deliver data to the correct destination and recipient only.

Accuracy. Data must be delivered exactly as sent. Corrupted data that cannot be recovered is useless.

Timeliness. Data must arrive on time. Real-time transmission (for video and audio) requires delivery as data is produced, without significant delay.

Jitter. This refers to variation in packet arrival time. If video packets sent every 30ms arrive at varying intervals (some at 30ms, others at 40ms), the resulting video quality becomes uneven and unwatchable.

1.1 Components of Data Communication

A data communications system comprises five distinct components that work together to ensure successful information transfer. Understanding each component and its role is essential for grasping how networks function at a fundamental level.

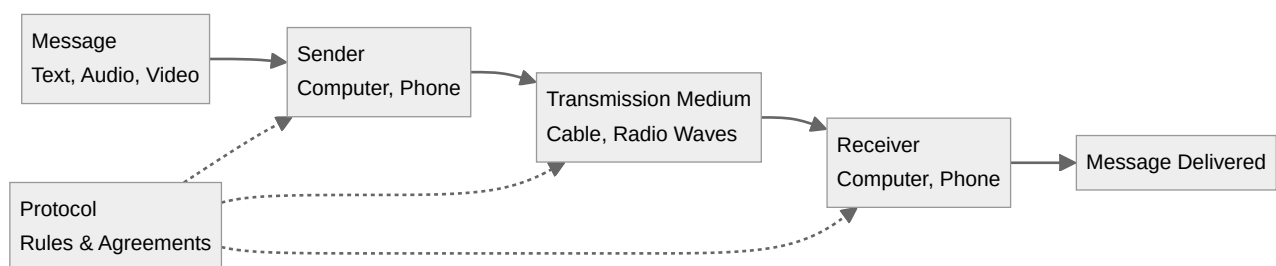


Figure 1.1 The five components of a data communication system

Table 1.1 Five Components of Data Communication

Component	Description	Examples
Message	The information to be communicated	Text, numbers, pictures, audio, video
Sender	The device that sends the data	Computer, phone, camera, workstation
Receiver	The device that receives the message	Computer, phone, printer
Transmission Medium	The physical path the message travels on	Twisted-pair wire, fiber-optic cable, radio waves
Protocol	A set of rules governing communication	TCP/IP, HTTP, FTP

Analogy: A protocol is like a language agreement. If two devices are connected but have no protocol, it is like a French speaker trying to communicate with a Japanese speaker. They can physically hear each other, but they cannot understand one another.

1.1.1 Data Representation

Information exists in various forms, and computers represent each form differently. All data eventually becomes bit patterns (sequences of 0s and 1s) for transmission.

Text. Text is represented as bit patterns. The prevalent coding system today is **Unicode**, which uses 32 bits to represent symbols from virtually any language. **ASCII** is an older system that now forms the first 127 characters of Unicode, providing backward compatibility with legacy systems.

Numbers. Unlike text which requires coding systems, numbers are directly converted to binary. This direct conversion makes mathematical operations simpler and more efficient.

Images. Images consist of a matrix of pixels (picture elements). Higher pixel counts yield better resolution but require more memory. A black-and-white image uses 1 bit per pixel, while grayscale and color images use more bits. Color images typically use the **RGB** method, combining Red, Green, and Blue components to produce the full spectrum of visible colors.

Audio. Audio represents a recording of sound. Unlike text or numbers which are discrete, audio is continuous and analog in nature. It must be sampled and converted to digital form for computer processing.

Video. Video represents recording of pictures or movies. It can be continuous (as from a television camera) or a combination of discrete images arranged sequentially to create the illusion of motion.

1.1.2 Data Flow (Communication Modes)

Communication between devices can occur in three distinct modes, each with specific characteristics and use cases. The choice of mode depends on the application's requirements for directionality and simultaneity.

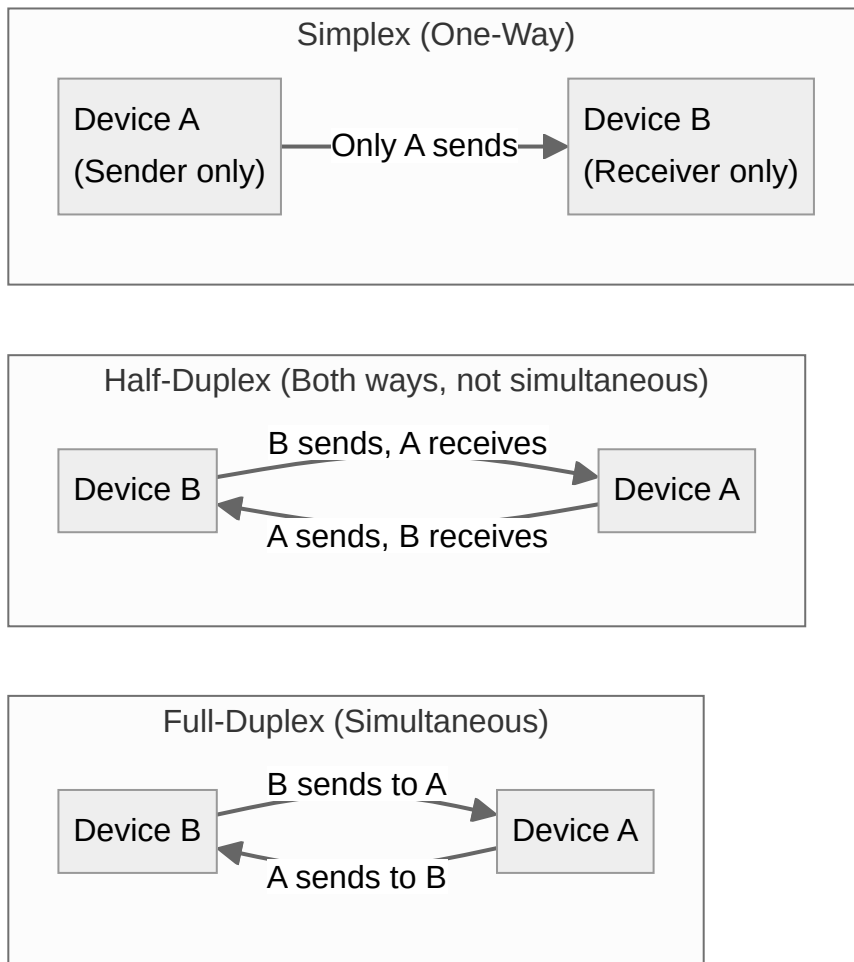


Figure 1.2 The three modes of data flow

Table 1.2 Comparison of Data Flow Modes

Mode	Direction	Analogy	Example
Simplex	Unidirectional: only one device transmits, the other only receives	A one-way street where traffic flows in a single direction only	Keyboard to monitor; radio broadcast
Half-Duplex	Bidirectional, but not simultaneous: devices take turns	A one-lane road allowing traffic in both directions, but only one direction at a time	Walkie-talkies (must say "Over" to switch)
Full-Duplex	Bidirectional and simultaneous: both devices transmit and receive at the same time	A two-way street with traffic flowing in both directions simultaneously	Telephone network

1.2 Networks

A network is the interconnection of a set of devices capable of communication. Two key terms define the participants in a network: a **host** (or end system) refers to devices like computers, desktops, laptops, phones, or security systems; while a **connecting device** refers to hardware that links the network together, such as routers (which connect networks to other networks), switches (which connect devices together), and modems (which change the form of data between analog and digital).

1.2.1 Network Criteria

For a network to be effective, it must satisfy three key criteria. First, **performance** is measured by transit time (the time for data to travel from source to destination) and response time (the time between a request and its response). An important trade-off exists between throughput and delay: sending more data increases throughput but may cause congestion, which in turn increases delay. Second, **reliability** is measured by the frequency of failure, recovery time after failure, and the network's robustness during catastrophic events. Third, **security** involves protecting data from unauthorized access, preventing damage to data, and ensuring against data loss.

1.2.2 Physical Structures

Type of Connection

A link represents the communications pathway that transfers data between devices. Two fundamental connection types exist.

Point-to-Point connection. A dedicated link connects exactly two devices, with the entire capacity of the link reserved for their exclusive use. This is analogous to a television remote: when you change the channel, you establish a direct line between the remote and the TV that no one else shares.

Multipoint (Multidrop) connection. More than two specific devices share a single link. The capacity is shared among multiple devices, either spatially (everyone communicates simultaneously, requiring sufficient capacity) or temporally (devices take turns). This resembles a conference call where multiple participants share the same line.

Physical Topology

Topology refers to the geometric representation of the relationship between devices (called nodes) and their connecting links. Four fundamental topologies form the basis of all network designs.

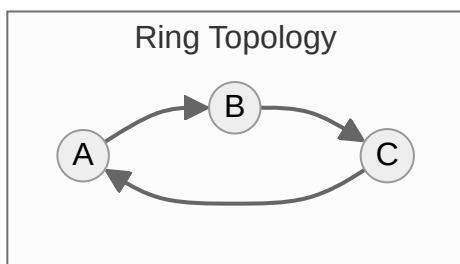
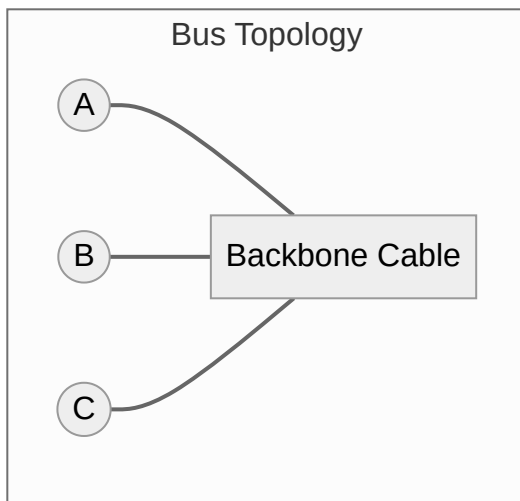
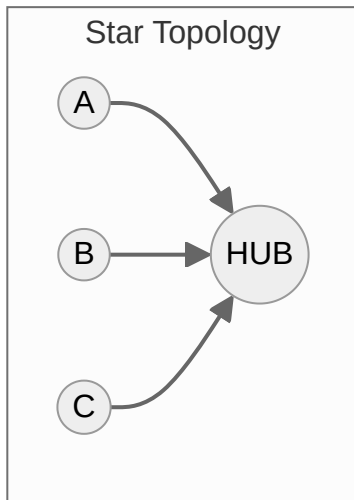
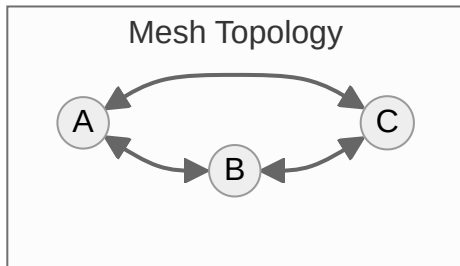


Figure 1.3 *The four fundamental network topologies*

Table 1.3 Comparison of Network Topologies

Topology	Structure	Advantages	Disadvantages	Use Case
Mesh	Every device has a dedicated point-to-point link to every other device. Formula: $n(n - 1)/2$ duplex links for n nodes.	No traffic problems; robust (if one link fails, others work); privacy guaranteed.	Expensive and bulky (massive cabling); difficult installation.	Regional telephone offices
Star	Each device connects only to a central hub. Devices do not link directly.	Cheaper than mesh (less cable); robust (one link failure affects only that device); easy to manage.	Single point of failure: if the hub fails, the entire network fails.	Local-Area Networks (LANs)
Bus	Multipoint configuration with one long backbone cable linking all devices via drop lines and taps.	Easiest to install; uses the least amount of cable.	Signal reflection at taps; fragile (a break stops all transmission).	Legacy LANs
Ring	Each device connects to two immediate neighbors. Signal travels in one direction; each device acts as a repeater.	Easy to install; simple fault isolation (device sounds alarm if signal not received).	Unidirectional; a break can disable the entire network (unless dual ring is used).	IBM Token Ring (legacy)

1.3 Network Types and Switching

1.3.1 Local Area Network (LAN)

A LAN is a privately owned network connecting hosts within a single office, building, or campus. Each host in a LAN has a unique address that identifies it on the network. LAN technology has evolved significantly over time.

In the past, all hosts on a LAN shared a common cable. If Host A sent a message to Host B, every host received the message, but only Host B would keep it. This approach was inefficient because bandwidth was wasted delivering messages to hosts that did not need them.

Modern LANs use a **switch**, which is an intelligent device that recognizes the destination address and guides packets only to the specific host that needs them. This reduces unnecessary traffic and allows multiple simultaneous conversations between different pairs of hosts.

1.3.2 Wide Area Network (WAN)

A WAN spans a wider geographical area, potentially covering a town, state, country, or even the entire world. Three key differences distinguish WANs from LANs: scope (LAN covers a building, WAN covers the world), connectivity (LAN connects hosts directly, WAN connects connecting devices like switches and routers), and ownership (LANs are privately owned while WANs are typically leased from communication companies).

Two types of WANs exist. A **point-to-point WAN** connects exactly two devices through a transmission medium. A **switched WAN** is a network with more than two ends, used in the global communication backbone, and is essentially a combination of several point-to-point WANs connected by switches.

Internetwork

It is rare to see a LAN or WAN in isolation. When two or more networks connect, they form an **internetwork** (or *internet* with a lowercase 'i'). For example, an office on the East Coast (LAN) might connect to an office on the West Coast (LAN) using a leased service (WAN). The router plays a critical role here: if you send a message to a colleague in the same office, the router blocks it and keeps it local; if you send it to the other office, the router passes it to the WAN.

1.3.3 Switching

An internet is a switched network, where a switch's primary job is to forward data from one network to another. Two fundamental switching approaches exist.

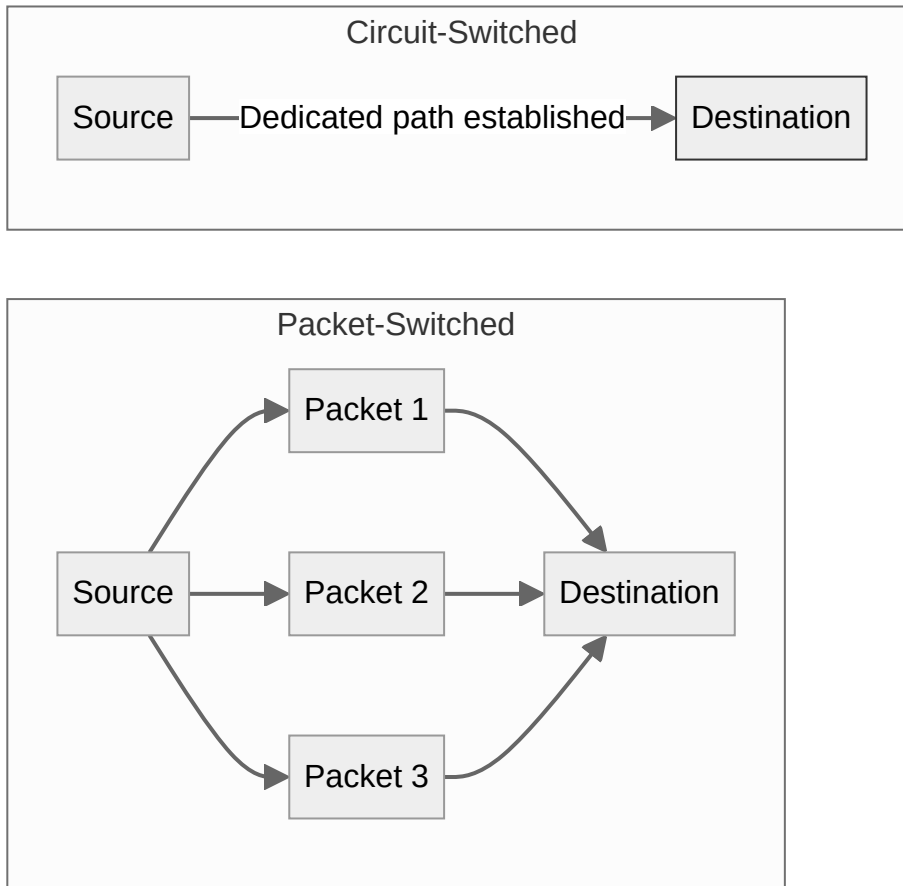


Figure 1.4 *Circuit switching vs. Packet switching*

Circuit-Switched Network. A dedicated connection (circuit) is established between two ends and remains reserved for the duration of the communication. This is analogous to a traditional telephone call: when you call someone, a specific line is reserved for you. Even if you stop talking, that line remains yours and no one else can use it. This approach guarantees a connection but is inefficient because unused capacity goes to waste.

Packet-Switched Network. Communication occurs in independent blocks of data called packets. This resembles postal mail: you do not reserve the entire road to send a letter; instead, you drop a letter (packet) in a box and it travels independently along with everyone else's mail. Routers use a **store-and-forward** mechanism, holding packets in a queue and sending them when the line becomes free. This approach is more efficient because lines are shared, though packets may encounter delays while waiting in queues.

1.3.4 The Internet and Its Hierarchy

The distinction between *internet* (lowercase, meaning two or more networks communicating) and *Internet* (uppercase, referring to the specific global collaboration of thousands of interconnected networks) is important. The Internet operates as a hierarchy with three levels: **backbones** (international ISPs like Sprint and AT&T, connected by peering points), **provider networks** (national or regional ISPs that pay to use

1.3.5 Accessing the Internet

Users must physically connect to an ISP, usually via a point-to-point WAN. Four common access methods exist: telephone networks (dial-up using a modem, or DSL which allows simultaneous voice and data), cable networks (using cable TV infrastructure, though speed varies based on neighbor usage), wireless networks (access via wireless WANs), and direct connections (large organizations leasing high-speed WANs directly from carriers).

Chapter 2: Network Models

Network models provide a structured framework for understanding how data moves through communication systems. While a simple protocol suffices for basic communication, complex systems like the Internet require dividing tasks into different layers, each with specific responsibilities. This layered approach, called protocol layering, is the architectural foundation of all modern networks.

2.1 Protocol Layering

To understand why layering is necessary, consider two scenarios involving friends Maria and Ann communicating with each other.

Scenario 1: Simple Face-to-Face Communication. When Maria and Ann are neighbors talking directly, communication occurs in a single layer. The protocol is straightforward: greet when meeting, use a common vocabulary, take turns speaking, and exchange parting words. No intermediaries are needed.

Scenario 2: Complex Remote Communication with Encryption. When Ann moves to a distant city, they must communicate via mail while keeping their ideas secret. This requires three layers: Layer 3 (High Level) where Maria speaks to a machine that writes plaintext; Layer 2 (Security) where the machine encrypts plaintext into ciphertext; and Layer 1 (Transmission) where the machine places the ciphertext in an envelope and mails it. At the receiving end, the process reverses: receive mail, decrypt, then read.

2.1.1 Why Use Layering?

Modularity through Independent Layers. A layer functions as a "black box": we care about its inputs and outputs, not its internal implementation. If Maria and Ann want to change their encryption method (Layer 2), they simply replace the Layer 2 machine. Layer 1 (mail) and Layer 3 (writing) remain completely untouched. This separation of service from implementation means you can change how a layer works without breaking the system, as long as the service it provides remains the same.

Intermediate Systems. In complex networks like the Internet, intermediate devices such as routers do not need to understand all layers. They only need the lower layers to pass data along, making the system cheaper and more efficient.

EXAM TIP

The main disadvantage of layering is that a single super-machine would theoretically be faster and easier than multiple interacting machines. However, if that super-machine breaks or needs updating, the entire system must be replaced, which is costly. Layering trades some efficiency for maintainability and flexibility.

2.1.2 Principles of Protocol Layering

When designing layered protocols, two fundamental principles must be followed.

First Principle: Bidirectional Capability. If communication is two-way, every layer must be able to perform opposite tasks. Layer 3 must both listen and talk; Layer 2 must both encrypt and decrypt; Layer 1 must both send and receive.

Second Principle: Identical Objects. The object passing between Layer X at the sender and Layer X at the receiver must be identical. Layer 3 at both sites handles plaintext; Layer 2 at both sites handles ciphertext; Layer 1 at both sites handles the physical mail object.

2.1.3 Logical Connections

Because of the principles above, we can imagine a logical connection directly between corresponding layers at sender and receiver. The physical flow of data moves vertically: down from Layer 3 to Layer 1 at the sender, across the physical medium, then up from Layer 1 to Layer 3 at the receiver. However, logically, we pretend there is a direct horizontal line between Layer 2 at the sender and Layer 2 at the receiver. The encryption layer does not care about envelopes or mail carriers; it effectively "talks" directly to the decryption layer at the other end.

2.2 TCP/IP Protocol Suite

The TCP/IP (Transmission Control Protocol/Internet Protocol) suite represents the standard protocol suite used for the Internet today. It is hierarchical, meaning each upper-level protocol relies on services provided by lower-level protocols. While originally defined as a 4-layer model, TCP/IP is today viewed as a 5-layer model for better understanding and teaching.

2.2.1 Layered Architecture: Who Uses Which Layers?

Different network devices operate at different depths within the layer stack. Understanding which devices process which layers is fundamental to network design and troubleshooting.

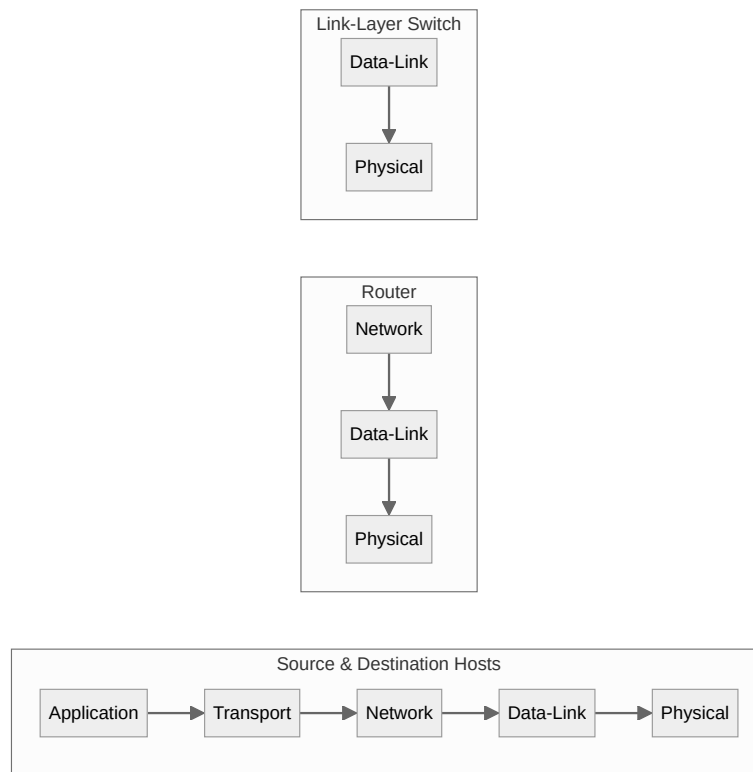


Figure 2.2 *Devices and the layers they process in the TCP/IP model*

2.2.2 The Five Layers of TCP/IP

The TCP/IP protocol suite is discussed from bottom (hardware) to top (user application).

Physical Layer. The Physical layer is responsible for carrying individual bits across the link. It connects devices via a transmission medium (cable or air), converting bits into signals (electrical or optical) that the medium can carry. The logical unit at this layer is a single bit.

Data-Link Layer. The Data-Link layer moves the packet (called a **frame** at this layer) across a specific link. It ensures hop-to-hop delivery from one node to the very next node, such as from your laptop to the Wi-Fi router. TCP/IP does not define a specific Data-Link protocol; it supports all standard protocols including Ethernet and Wi-Fi.

Network Layer. The Network layer is responsible for host-to-host communication and routing. The packet at this layer is called a **datagram**. The key protocol is IP (Internet Protocol), which is connectionless and provides no flow or error control, relying on higher layers for reliability. Auxiliary protocols include ICMP (which reports routing problems), DHCP (which assigns IP addresses), and ARP (which finds the link-layer MAC address when given a network-layer IP address). The Network layer functions like a postal sorting center: it examines the destination address and determines which route the packet should take next.

Transport Layer. The Transport layer handles process-to-process delivery, separating network duties from application duties. It creates an end-to-end logical connection where the Transport layer at the source "talks" directly to the Transport layer at the destination. Two key protocols exist: **TCP** (Transmission Control Protocol) is connection-oriented and reliable, providing flow control and error correction; **UDP** (User Datagram Protocol) is connectionless, unreliable, but fast, making it suitable for small messages where retransmission would be too slow. Think of TCP as certified mail with a receipt and delivery guarantee, and UDP as regular mail that is faster and cheaper but offers no delivery confirmation.

Application Layer. The Application layer enables the user (human or software) to access the network. Communication occurs between two processes (running programs). Common protocols include HTTP for the World Wide Web, SMTP for email, FTP for file transfer, and DNS for translating domain names into IP addresses.

2.2.3 Encapsulation and Decapsulation

Encapsulation and decapsulation describe how data is packaged and unpackaged as it moves vertically through the layers. This process is fundamental to understanding how data travels through a network.



Figure 2.3 *The encapsulation and decapsulation process*

At the **source host**, data moves downward through the layers. Each layer treats data from the layer above as payload and adds its own header. The Application layer creates the message. The Transport layer takes the message as payload and adds a transport header containing source and destination port numbers along with flow control information, producing a segment (TCP) or user datagram (UDP). The Network layer takes the segment as payload and adds a network header with source and destination IP addresses, producing a datagram. The Data-Link layer takes the datagram as payload and adds a link-layer header with source and destination MAC addresses, producing a frame.

At a **router**, the device decapsulates by receiving the frame and removing the header to read the datagram. It checks the IP address to find the best path (routing), generally without changing the datagram content. Then it encapsulates by wrapping the datagram into a new frame suitable for the next link.

At the **destination host**, data moves upward. Each layer removes its specific header, checks for errors, and hands the pure payload to the layer above until the Application layer receives the original message.

Analogy (Russian Nesting Doll): Think of encapsulation like nested containers. The innermost doll is the letter (Message). It goes inside an envelope with apartment numbers (Port Numbers), which goes inside a mailbag with the building address (IP Address), which goes inside a delivery truck container (MAC Address).

2.2.4 Addressing in TCP/IP

Communication requires identifying who is talking to whom. Different layers use different address types, each serving a specific purpose in the delivery process.

Table 2.2 Addressing at Each TCP/IP Layer

Layer	Packet Name	Address Type	Scope	Example
Application	Message	Names (user-specific)	Global	email@domain.com, google.com
Transport	Segment/Datagram	Port Numbers (identifies the program)	Local	Port 80 (Web), Port 25 (Email)
Network	Datagram	Logical Addresses (IP)	Global (identifies the connection)	192.168.1.1
Data-Link	Frame	Physical Addresses (MAC)	Local (identifies the device on the link)	00:1A:2B:3C:4D:5E
Physical	Bits	None	N/A	Bits do not need addresses

2.2.5 Multiplexing and Demultiplexing

Because the TCP/IP suite allows multiple protocols to run simultaneously, mechanisms are needed to mix and un-mix data traffic.

Multiplexing occurs at the source: a protocol at a lower layer accepts packets from several higher-layer protocols. For example, the IP protocol at the Network layer can carry both TCP and UDP packets simultaneously.

Demultiplexing occurs at the destination: a protocol decapsulates a packet and delivers it to the correct higher-layer protocol. When IP receives a packet, it examines a protocol field in the header to determine whether the packet should go to TCP or UDP.

2.3 The OSI Model

While TCP/IP serves as the practical standard for the Internet, the International Organization for Standardization (ISO) established the Open Systems Interconnection (OSI) model in the late 1970s as a comprehensive reference standard. An **open system** is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture. The OSI model is not itself a protocol but rather a blueprint for understanding and designing flexible, interoperable network architectures.

2.3.1 OSI versus TCP/IP

When comparing the two models, a distinct difference emerges in how they organize layers. The TCP/IP suite lacks distinct Session and Presentation layers. Instead, TCP/IP combines the duties of Session, Presentation, and Application layers into a single Application Layer.

TCP/IP combined these layers for good reasons. Some Session layer functions, like tracking connections, are already handled by the Transport layer (TCP). The Application layer in TCP/IP is not monolithic: if a specific application needs session or presentation features like encryption or dialogue control, those features can be built directly into that application's software rather than requiring separate, strict layers. This design offers greater flexibility.

2.3.2 Why OSI Did Not Succeed

Originally, experts believed OSI would replace TCP/IP, but this did not happen. Three main factors contributed to OSI's failure to gain dominance. First, **timing and cost**: TCP/IP was already fully deployed, and changing everything to OSI would have been prohibitively expensive and time-consuming. Second, **undefined layers**: the Session and Presentation layers were never fully defined. Their services were listed conceptually, but the actual protocols and software to implement them were never fully developed. Third, **performance**: initial implementations of OSI did not demonstrate high enough performance to convince Internet authorities to abandon the already-functional TCP/IP suite.

Chapter 3: Data and Signals

While previous chapters discussed "logical" communication between applications, at the Physical Layer communication becomes physical. We do not send files or pictures through wires; we send signals. The transmission medium (cable or air) carries electromagnetic signals, so data must be transformed into these signals for transmission.

3.1 Analog and Digital Signals

Both data (the information being conveyed) and signals (the transmission mechanism) can be either analog or digital. Understanding this distinction is fundamental to all of data communications.

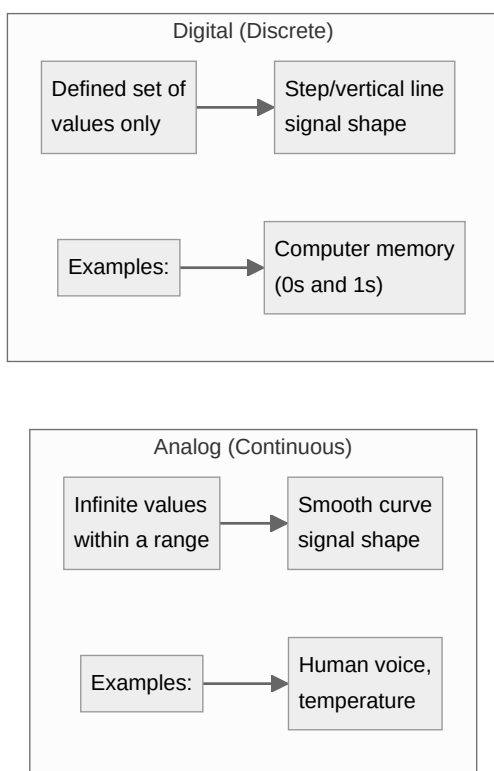


Figure 3.1 *Analog vs. Digital: continuous vs. discrete representation*

Analog signals are continuous, taking on infinite values within a range. An analog clock illustrates this concept perfectly: the hands move continuously, flowing smoothly from 8:05 to 8:06, passing through every microsecond in between. Human voice and temperature readings are naturally analog.

Digital signals are discrete, limited to a defined set of values. A digital clock demonstrates this: the display jumps suddenly from 8:05 to 8:06 with no "in-between" state shown. Computer memory, represented as 0s and 1s, is fundamentally digital.

repeating pattern. In data communications, we typically use periodic analog signals (like sine waves for carrier signals) and nonperiodic digital signals (for actual data transmission).

3.2 Periodic Analog Signals

Periodic analog signals are classified into two types: simple signals (a single sine wave that cannot be broken down further) and composite signals (made of multiple sine waves combined). The sine wave is the most fundamental periodic analog signal, appearing as a smooth, oscillating curve.

3.2.1 Sine Wave Parameters

A sine wave is completely described by three parameters: peak amplitude, frequency, and phase.

Peak Amplitude. The peak amplitude is the absolute value of the signal's highest intensity, proportional to the energy the signal carries. For electric signals, it is measured in volts. The voltage in US homes is stated as 110-120 V, but this is the RMS (Root Mean Square) value; the actual peak amplitude is higher (approximately 155-170 V). The relationship is:

Peak Amplitude from RMS

$$\text{Peak} = \sqrt{2} \times \text{RMS}$$

A battery provides constant voltage (e.g., 1.5 V), which is technically a sine wave with zero frequency and constant amplitude.

Period and Frequency. Period and frequency describe the same characteristic from different perspectives. The **period (T)** is the time in seconds required to complete one cycle. The **frequency (f)** is the number of periods (cycles) in one second. They are inverses of each other.

Fundamental Relationship

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

Period is formally expressed in seconds (s), while frequency is expressed in Hertz (Hz), meaning cycles per second. Frequency also measures the rate of change: high frequency means change happens over a short span of time, while low frequency means change happens over a long span.

Example: House Power Frequency to Period

Scenario: The power used in US homes has a frequency of 60 Hz. What is the period?

Solution:

$$T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ s} = 16.6 \text{ ms}$$

The period is 16.6 ms. Our eyes cannot detect a light bulb flickering at this speed, so it appears constant.

Example: Time Conversion

Task: Express a period of 100 ms in microseconds (μs).

Solution: $100 \text{ ms} = 100 \times 1000 \mu\text{s} = 100,000 \mu\text{s}$

Frequency Unit Conversions**Table 3.1 Frequency Unit Prefixes**

Unit	Equivalent	Unit	Equivalent
1 Hz	1 cycle/second	1 kHz (kilohertz)	$10^3 \text{ Hz} = 1,000 \text{ Hz}$
1 MHz (megahertz)	$10^6 \text{ Hz} = 1,000,000 \text{ Hz}$	1 GHz (gigahertz)	10^9 Hz
1 THz (terahertz)	10^{12} Hz		

Two Extremes of Frequency

Zero Frequency (0 Hz): The signal does not change at all, maintaining a constant voltage level forever. It never completes a cycle. Example: a battery providing DC current.

Infinite Frequency (∞ Hz): The signal changes instantaneously, jumping from one level to another in zero time. Since $f = 1/T$, if $T = 0$, then $f = \infty$. This is theoretical, but it helps describe sudden spikes or digital transitions.

3.2.2 Phase

Phase (or phase shift) describes the position of the waveform relative to time zero ($t = 0$). It indicates the status of the first cycle, essentially showing where in the cycle the wave starts when the clock begins.

Table 3.2 Phase Shift Descriptions

Phase	Starting Position	Visual Description
0°	Starts at 0 amplitude, increases	Normal, unshifted wave
90°	Starts at peak amplitude, decreases	Shifted left by 1/4 cycle
180°	Starts at 0 amplitude, decreases (goes negative)	Shifted left by 1/2 cycle

Phase Calculation

$$\text{Phase} = \frac{\text{Offset in cycles}}{1 \text{ cycle}} \times 360^\circ$$

Example: Phase Calculation

Problem: A sine wave is offset 1/6 cycle with respect to time 0. What is the phase?

Solution:

$$\text{Phase} = \frac{1/6}{1} \times 360^\circ = 60^\circ$$

3.2.3 Wavelength

Wavelength (λ) characterizes a signal traveling through a transmission medium. It binds the period (T) or frequency (f) to the propagation speed of the medium. Wavelength is defined as the distance a simple signal can travel in one period.

Wavelength Formula

$$\lambda = c \times T = \frac{c}{f}$$

where c is the propagation speed of the medium

Frequency is independent of the medium (a 60 Hz signal is 60 Hz whether traveling through copper or air), but wavelength depends on both frequency and medium because propagation speed changes across different media. In a vacuum, light travels at 3×10^8 m/s, while in a cable the speed is slower, producing a shorter wavelength.

3.2.4 Time and Frequency Domains

To fully understand a signal, we can view it from two different perspectives. The **time-domain plot** shows amplitude versus time, illustrating how the signal changes moment by moment. Its limitation is that phase is not explicitly shown, and reading frequency from a wavy line is difficult. The **frequency-domain plot** shows peak amplitude versus frequency, revealing the "ingredients" of the signal. A complete sine wave appears as a single vertical spike whose position indicates frequency and whose height indicates peak amplitude. The frequency domain is more compact and allows immediate reading of values without tracing curves.

When multiple sine waves are mixed together, the time domain shows a messy, complex waveform, while the frequency domain displays clean, distinct spikes that make analysis much easier.

3.2.5 Composite Signals and Fourier Analysis

While simple sine waves are useful for power delivery or alarm signals, they cannot carry data. If you sent a single sine wave over a phone line, the receiver would hear only a continuous buzz. To convey meaningful information (like a conversation), we need variety, which is achieved through **composite signals** made of many simple sine waves combined.

In the 1900s, Jean-Baptiste Fourier proved a critical concept: *any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases*. The decomposition depends on the signal type:

Table 3.3 Fourier Decomposition by Signal Type

Signal Type	Decomposition Result	Frequency Domain Appearance
Periodic composite	A series of discrete frequencies (integer multiples like $f, 2f, 3f$)	Distinct spikes (harmonics)
Nonperiodic composite	A combination of continuous frequencies (all real values)	A continuous curve

For example, a periodic square wave decomposes into a fundamental frequency (f) plus odd harmonics ($3f, 5f, 7f, \dots$) with progressively decreasing amplitudes. A human voice is nonperiodic and contains an infinite number of frequencies between 0 and 4 kHz, appearing as a continuous curve in the frequency domain.

3.2.6 Bandwidth

The **bandwidth** of a composite signal is the range of frequencies it contains, calculated as the difference between the highest and lowest frequencies.

Bandwidth Formula

$$B = f_h - f_l$$

where f_h is the highest frequency and f_l is the lowest frequency

For a periodic signal, the bandwidth includes all discrete frequencies (spikes) between the minimum and maximum. For a nonperiodic signal, the bandwidth encompasses the entire continuous range between minimum and maximum.

Example: Discrete Spikes Bandwidth

Scenario: A periodic signal has sine waves at frequencies 100, 300, 500, 700, and 900 Hz.

Calculation: $B = 900 - 100 = 800$ Hz

The spectrum shows exactly 5 spikes at these frequencies.

Example: Finding the Lowest Frequency

Scenario: Bandwidth is 20 Hz. Highest frequency is 60 Hz. What is the lowest?

Solution: $f_l = f_h - B = 60 - 20 = 40$ Hz

Example: Nonperiodic Signal Bandwidth

Scenario: Bandwidth = 200 kHz. Middle frequency = 140 kHz.

Logic: The bandwidth extends 100 kHz above and 100 kHz below the middle.

$$f_l = 140 - 100 = 40 \text{ kHz}$$

$$f_h = 140 + 100 = 240 \text{ kHz}$$

Table 3.4 Real-World Bandwidth Examples

Application	Bandwidth	Total Frequency Range
AM Radio (per station)	10 kHz	530 to 1700 kHz
FM Radio (per station)	200 kHz (higher quality than AM)	88 to 108 MHz
Analog TV (B&W video)	4 MHz	Varies by channel (6 MHz per channel)

3.3 Digital Signals

Information can be represented by a digital signal with discrete values instead of an analog signal. The basic form uses two levels: a positive voltage represents 1 and zero voltage represents 0. However, digital signals can also have more than two levels, allowing us to send more than 1 bit per level.

Bits per Level

$$\text{Bits per level} = \log_2(L)$$

where L is the number of signal levels

Table 3.5 Bits per Level Examples

Levels (L)	Bits per Level	Notes
2	$\log_2(2) = 1$ bit	Basic binary (0 or 1)
4	$\log_2(4) = 2$ bits	Can represent 00, 01, 10, 11
8	$\log_2(8) = 3$ bits	Can represent 000 through 111
9	$\log_2(9) \approx 3.17$	Must round up to 4 bits (which can support up to 16 levels)

3.3.1 Bit Rate

Since digital signals are rarely periodic, we do not use frequency to describe them. Instead, we use **bit rate**, defined as the number of bits sent in one second, measured in bits per second (bps).

Example: Text Download Bit Rate

Scenario: Downloading 100 pages per second. Each page = 24 lines x 80 characters. Each character = 8 bits.

Calculation:

$$\text{Bit Rate} = 100 \times 24 \times 80 \times 8 = 1,536,000 \text{ bps} = 1.536 \text{ Mbps}$$

Example: Digitized Voice

Scenario: 4 kHz analog voice, sampled at twice the frequency (2×4000 Hz), using 8 bits per sample.

Calculation:

$$\text{Bit Rate} = 2 \times 4000 \times 8 = 64,000 \text{ bps} = 64 \text{ kbps}$$

Example: HDTV Bit Rate

Scenario: 1920 x 1080 pixels, refreshed 30 times per second, 24 bits per color pixel.

Calculation:

$$\text{Bit Rate} = 1920 \times 1080 \times 30 \times 24 \approx 1.5 \text{ Gbps}$$

This is enormous. TV stations use compression to reduce this to 20-40 Mbps for transmission.

3.3.2 Bit Length

Just as analog signals have wavelength (the distance of one cycle), digital signals have bit length, defined as the distance one bit occupies on the transmission medium.

Bit Length Formula

$$\text{Bit Length} = \text{Propagation Speed} \times \text{Bit Duration}$$

3.3.3 Digital Signal as a Composite Analog Signal

According to Fourier analysis, a digital signal is actually a composite analog signal with infinite bandwidth. This may seem surprising, but the reasoning is clear: a digital signal has vertical lines representing instant jumps from one level to another. A vertical line implies a frequency of infinity (instantaneous change), while horizontal segments imply zero frequency (no change). To create this shape theoretically requires all frequencies between 0 and infinity.

3.3.4 Transmission of Digital Signals

Since a digital signal theoretically requires infinite bandwidth, we need practical approaches to transmit it. Two fundamental approaches exist: baseband and broadband transmission.

Baseband Transmission. Baseband sends a digital signal over a channel without converting it to an analog signal (no modulation). It requires a **low-pass channel** (a channel where bandwidth starts from 0 Hz). When the low-pass channel has wide bandwidth (dedicated medium like coaxial cable or fiber in a LAN), the signal shape is preserved accurately. However, when bandwidth is limited, we must approximate: the signal will look curved rather than perfectly square. The harmonics rule applies: a rough approximation uses bandwidth $B = N/2$ (first harmonic only), producing a sine-like wave; better approximation uses $B = 3N/2$ or $5N/2$ (including third or fifth harmonics), producing a squarer waveform. The key takeaway: required bandwidth is proportional to bit rate.

Baseband Bandwidth Requirements (for bit rate N)

$$\text{Rough (minimum): } B = \frac{N}{2} \quad \text{Better: } B = \frac{3N}{2} \quad \text{Best: } B = \frac{5N}{2}$$

Example: Required Bandwidth for 1 Mbps

Goal: Send 1 Mbps (1,000,000 bps).

Solutions:

Minimum (rough): $B = 1 \text{ Mbps} / 2 = 500 \text{ kHz}$

Better (1st + 3rd harmonics): $B = 3 \times 500 = 1.5 \text{ MHz}$

Best (1st + 3rd + 5th harmonics): $B = 5 \times 500 = 2.5 \text{ MHz}$

Example: Maximum Bit Rate

Scenario: Low-pass channel with bandwidth of 100 kHz. What is the maximum speed?

Solution: Using rough approximation ($B = N/2$):

$$100 \text{ kHz} = \frac{N}{2} \quad \Rightarrow \quad N = 200 \text{ kbps}$$

Broadband Transmission (Modulation). When a low-pass channel is unavailable (bandwidth does not start at 0 Hz), we cannot use baseband. Instead, we use broadband transmission: converting the digital signal to an analog signal through **modulation**. The channel used is a **bandpass channel** (bandwidth starts at a non-zero frequency). The process involves taking a single-frequency analog signal called a **carrier** and modifying one of its properties (amplitude, frequency, or phase) to encode the digital data. The receiver then converts the analog signal back to digital through **demodulation**.

Real-World Example: Telephone Modem

Telephone lines were built for voice (analog) with a narrow bandwidth of 0 to 4 kHz. If treated as a low-pass channel for digital data, the maximum speed would be only 8 kbps. Instead, modems treat the line as a bandpass channel, converting the computer's digital signal into an analog signal that fits within the phone line's frequency range, achieving much higher speeds.

3.4 Transmission Impairment

Transmission media are not perfect. As signals travel, they suffer from impairment, meaning the received signal is rarely identical to the transmitted signal. Three primary causes of impairment exist: attenuation, distortion, and noise.

3.4.1 Attenuation

Attenuation refers to a loss of energy. When a signal travels through a medium, it must overcome resistance, and some electrical energy converts into heat (just as a wire warms when current passes through). The signal becomes weaker as distance increases. To compensate, we use **amplifiers** to boost the signal strength back to usable levels.

The Decibel (dB). Engineers use decibels to measure relative signal strength between two points.

Decibel Formulas

$$dB = 10 \log_{10} \left(\frac{P_2}{P_1} \right) \quad (\text{for power})$$

$$dB = 20 \log_{10} \left(\frac{V_2}{V_1} \right) \quad (\text{for voltage})$$

Decibel Key Rules

Negative dB: The signal is attenuated (weaker at the destination).

Positive dB: The signal is amplified (stronger at the destination).

Example: A signal travels through three segments with dB values of -3 dB, $+7$ dB, and -3 dB. Total: $dB = -3 + 7 - 3 = +1$ dB, meaning the signal has gained power overall.

3.4.2 Distortion

Distortion means the signal changes its form or shape. This typically occurs in composite signals made of different frequencies. Each frequency component travels at its own propagation speed through the medium, arriving at the destination with different delays. These different delays cause phase shifts, and when the components are recombined, the signal shape is deformed compared to what was sent.

Consider a sender transmitting a clean composite wave made of frequencies A, B, and C. In the medium, frequency B travels slightly slower than A. At the receiver, when the components recombine, the wave appears lumpy or flattened because component B arrived later than the others.

3.4.3 Noise

Noise is external energy that corrupts the signal by adding unwanted information. Four types of noise affect communication systems.

Table 3.6 Types of Noise

Type	Cause	Characteristics
Thermal Noise	Random motion of electrons in the wire	Creates background hum or static; always present
Induced Noise	External sources like motors or appliances	The motor acts as a transmitting antenna, the wire as receiving antenna
Crosstalk	Effect of one wire on another	Like hearing a faint conversation on an old phone line because wires are too close
Impulse Noise	High-energy spikes in very short time	Caused by lightning strikes, power surges; sudden and severe

3.4.4 Signal-to-Noise Ratio (SNR)

To understand the theoretical limit of data transmission speed, we measure channel quality using SNR, the ratio of average signal power to average noise power. It represents the ratio of what is wanted (signal) to what is not wanted (noise).

SNR Formulas

$$\text{SNR} = \frac{\text{Average Signal Power}}{\text{Average Noise Power}} \quad (\text{ratio})$$

$$\text{SNR}_{dB} = 10 \log_{10}(\text{SNR}) \quad (\text{in decibels})$$

Analogy: *High SNR* is like talking to a friend in a quiet library: your voice (signal) is clear and the background (noise) is near zero. *Low SNR* is like talking at a loud rock concert: the music (noise) drowns out your voice (signal), making communication difficult.

EXAM TIP

Do not confuse SNR (the raw ratio) with SNR_{dB} (the decibel version). If $\text{SNR} = 10,000$, then $\text{SNR}_{dB} = 40$ dB. Calculations for data limits typically require converting dB back to the standard ratio first.

3.5 Data Rate Limits

A critical question in data communications asks: "How fast can we send data?" The data rate (in bps) depends on three factors: the bandwidth available, the number of signal levels used, and the quality of the channel (noise level). Two theoretical formulas provide answers: Nyquist for ideal noiseless channels and Shannon for real noisy channels.

3.5.1 Nyquist Bit Rate (Noiseless Channel)

Assuming a perfectly noiseless channel, the maximum bit rate depends on bandwidth and the number of signal levels.

Nyquist Bit Rate Formula

$$\text{BitRate} = 2 \times \text{Bandwidth} \times \log_2(L)$$

where L = number of signal levels

Theoretically, any bit rate is achievable by increasing L . However, increasing levels imposes a burden on the receiver: distinguishing between 2 levels is easy (voltage is either 0 or 5 V), but distinguishing between 64 levels requires incredibly sophisticated receivers to tell the difference between adjacent levels like 32 and 33. Increasing levels reduces system reliability.

3.5.2 Shannon Capacity (Noisy Channel)

Noiseless channels do not exist in reality. In 1944, Claude Shannon introduced a formula for the absolute maximum data rate of a noisy channel.

Shannon Capacity Formula

$$C = \text{Bandwidth} \times \log_2(1 + \text{SNR})$$

where C = capacity in bps, and SNR is the raw ratio (not in dB)

Notice that signal levels (L) do not appear in this formula. No matter how many levels are used or how clever the encoding, the data rate cannot exceed the Shannon Capacity. This formula defines the characteristic of the channel itself, not the transmission method.

3.5.3 Using Both Limits in Practice

In practice, engineers use both methods together. First, Shannon's formula determines the upper limit (the "speed limit of the road"). Then, Nyquist's formula determines how many signal levels are needed to achieve that speed. If Shannon says the maximum is 10 Mbps and Nyquist says we need 8 levels to achieve it, we know both the theoretical limit and the practical implementation requirement.

Example: Applying Both Limits

Scenario: A channel has bandwidth 1 MHz and SNR = 20 dB.

Step 1: Convert SNR from dB to ratio:

$$20 = 10 \log_{10}(\text{SNR}) \Rightarrow \text{SNR} = 10^2 = 100$$

Step 2: Apply Shannon's formula:

$$C = 10^6 \times \log_2(1 + 100) = 10^6 \times \log_2(101) \approx 10^6 \times 6.66 = 6.66 \text{ Mbps}$$

Step 3: Use Nyquist to find levels needed:

$$6.66 \times 10^6 = 2 \times 10^6 \times \log_2(L) \Rightarrow \log_2(L) = 3.33 \Rightarrow L = 2^{3.33} \approx 10 \text{ levels}$$

Chapter 6: Bandwidth Utilization

Bandwidth represents one of the most precious resources in data communications. Rather than installing new physical cables every time additional capacity is needed, we employ multiplexing techniques that allow the simultaneous transmission of multiple signals across a single data link. Multiplexing is practical whenever the bandwidth of the medium (the pipe) exceeds the bandwidth needs of the individual devices connected to it. The goal is efficiency: maximizing the use of high-bandwidth media like fiber optics or microwaves by sharing them among many lower-speed users.

6.1 Multiplexing Techniques

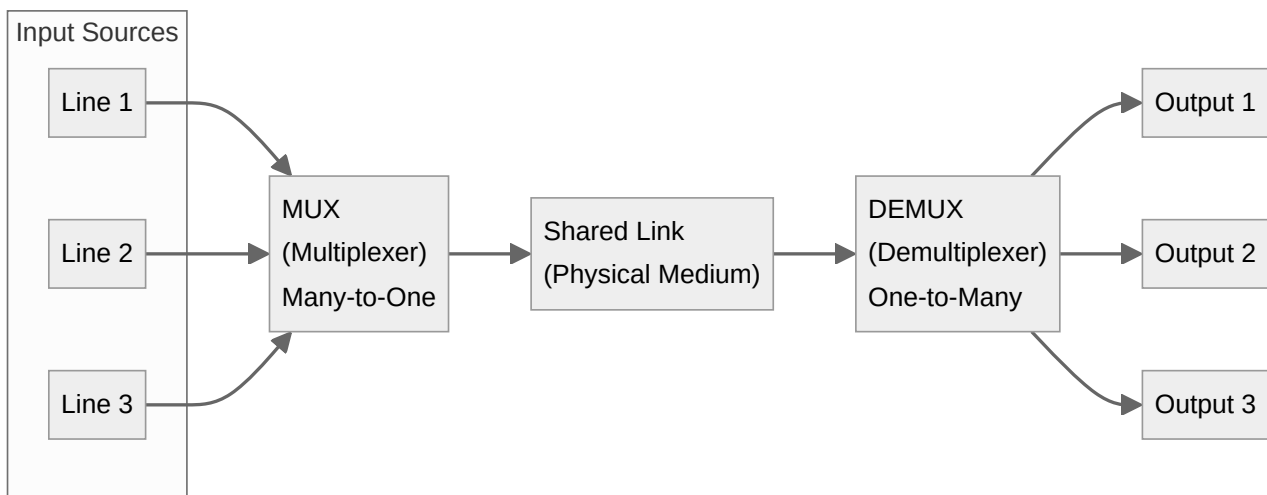


Figure 6.1 The multiplexing system: MUX combines inputs, DEMUX separates outputs

The multiplexing system consists of n input lines on the left, a **MUX (Multiplexer)** that combines the inputs into a single stream (many-to-one), a shared physical link (the wire), and a **DEMUX (Demultiplexer)** that separates the combined stream back into individual transmissions (one-to-many). It is important to distinguish between a **link** (the physical path, the cable itself) and a **channel** (the portion of a link carrying a specific transmission). One link can carry n channels simultaneously.

Three basic multiplexing techniques exist: **FDM (Frequency-Division Multiplexing)** for analog signals, **WDM (Wavelength-Division Multiplexing)** for analog light signals, and **TDM (Time-Division Multiplexing)** for digital signals.

6.1.1 Frequency-Division Multiplexing (FDM)

FDM is an analog technique used when the bandwidth of the link (in Hz) is wider than the combined bandwidths of the signals to be transmitted. Each sender's signal modulates a different carrier frequency.

Guard bands, which are strips of unused bandwidth, separate channels to prevent signals from overlapping or interfering with each other.

Analogy: FDM works like FM Radio stations. All stations transmit through the air simultaneously without interfering because Station A broadcasts at 98.5 MHz while Station B broadcasts at 101.5 MHz. Each occupies a different "channel" within the same medium.

The FDM Process. At the sender side (multiplexing), each source generates a signal, each signal modulates a different carrier frequency (f_1 , f_2 , f_3), the modulated signals are combined into one composite signal, and that composite signal is sent across the link. At the receiver side (demultiplexing), the demultiplexer receives the composite signal, uses filters to decompose it into its component parts, a demodulator separates each signal from its carrier frequency, and the original signals are delivered to their output lines.

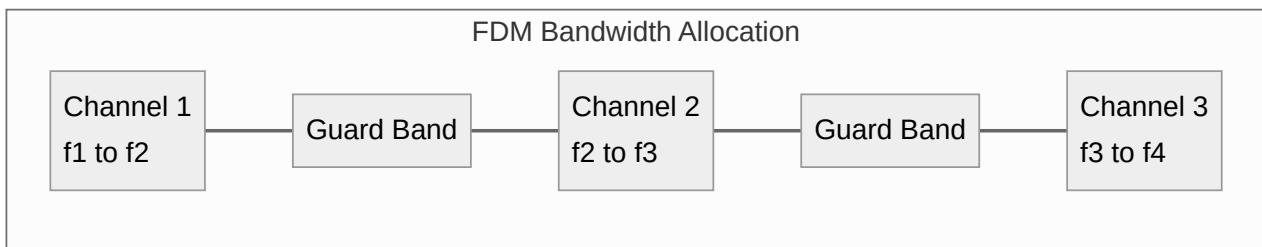


Figure 6.2 FDM divides bandwidth into channels separated by guard bands

Example: Voice Channels (No Guard Bands)

Scenario: Three voice channels, each occupying 4 kHz, combined onto one link with bandwidth 12 kHz (20 to 32 kHz). No guard bands.

Solution (Frequency Shift):

Channel 1: Shifted to 20-24 kHz

Channel 2: Shifted to 24-28 kHz

Channel 3: Shifted to 28-32 kHz

At Receiver: Filter 1 listens to 20-24 kHz, Filter 2 listens to 24-28 kHz, Filter 3 listens to 28-32 kHz.

The Analog Carrier System Hierarchy

Telephone companies use a hierarchical system to multiplex many low-bandwidth voice lines onto fewer high-bandwidth channels. FDM builds up from individual voice channels into massive "Jumbo Groups."

Table 6.1 Analog Carrier System Hierarchy

Level	Composition	Bandwidth
Voice Channel	1 voice line	4 kHz
Group	12 voice channels	48 kHz
Supergroup	5 groups (60 voice channels)	240 kHz
Master Group	10 supergroups (600 voice channels)	2.52 MHz
Jumbo Group	6 master groups (3600 voice channels)	16.984 MHz

Other FDM Applications

FDM serves as the backbone of broadcasting. AM radio stations each receive 10 kHz within the total band of 530 to 1700 kHz. FM radio offers higher quality with 200 kHz per station across 88 to 108 MHz. Television broadcasting assigns 6 MHz per channel. First-generation cellular phones (1G) used FDM, giving each user two 30-kHz channels (one for sending voice, one for receiving), with voice signals modulated using FM to occupy the full channel width.

Example: Cellular Capacity (AMPS)

Scenario: AMPS uses two 25 MHz bands: send (824-849 MHz) and receive (869-894 MHz). Each user requires 30 kHz.

Solution:

$$\text{Total channels} = \frac{25 \text{ MHz}}{30 \text{ kHz}} = \frac{25,000}{30} \approx 833 \text{ channels}$$

(In practice, the band is divided into 832 channels.)

42 channels are reserved for control signals. Available for users: $832 - 42 = 790$ channels.

6.1.2 Wavelength-Division Multiplexing (WDM)

WDM is designed specifically to utilize the immense data rate capability of fiber-optic cables. Using a single fiber for one signal wastes capacity; WDM combines several optical lines into one. Conceptually, WDM is identical to FDM: it combines signals of different frequencies. The difference is that WDM deals with extremely high frequencies (light waves).

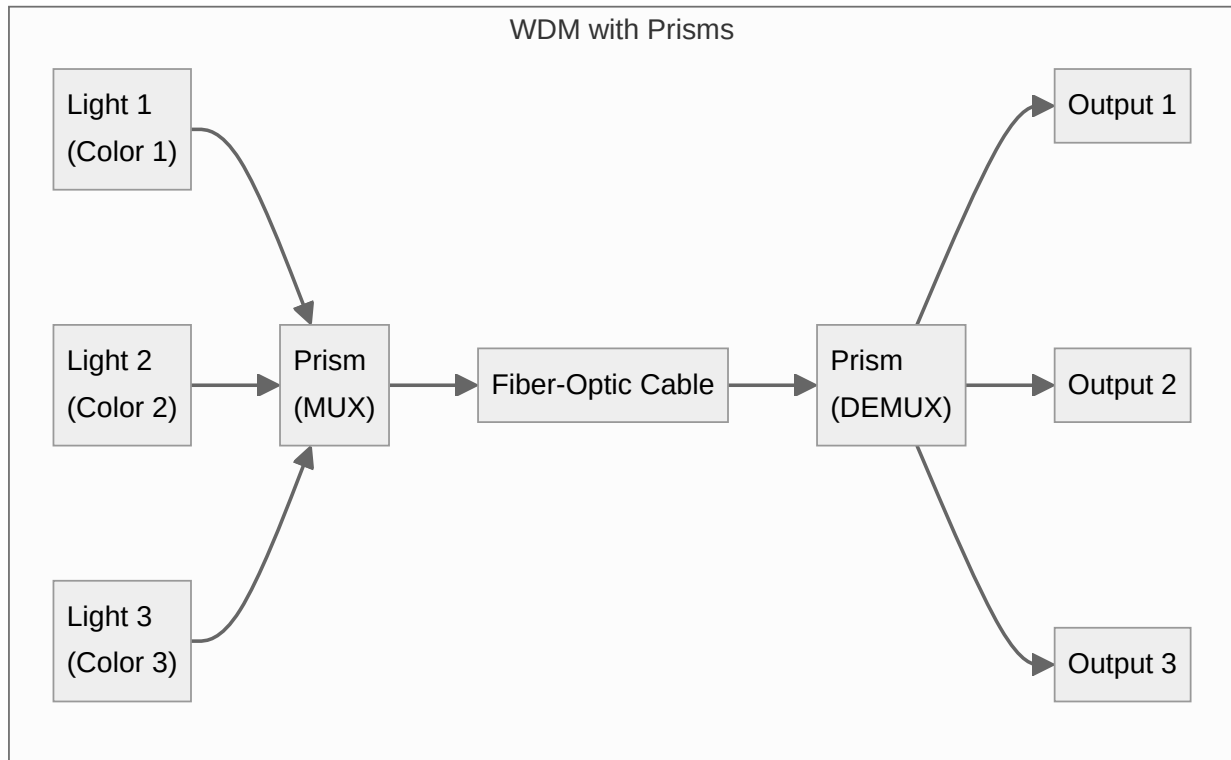


Figure 6.3 WDM uses prisms to combine and separate light signals

Although the technology is complex, the underlying physics is simple: a prism combines multiple input light beams (each a narrow band or color) into one wider beam at the multiplexer, and another prism separates the combined beam back into individual narrow beams at the demultiplexer. **Dense WDM (DWDM)** is a newer method that packs channels very close together for even greater efficiency.

6.1.3 Time-Division Multiplexing (TDM)

TDM is a digital process. Instead of sharing portions of bandwidth (like FDM/WDM), users share portions of time. The high bandwidth of the link functions as a single fast lane where users take turns occupying the link sequentially.

Synchronous TDM

In synchronous TDM, the multiplexer allocates the same slot to the same device at all times. Each input connection has an allotment in the output even if it is not sending data. This means if a device is silent, its time slot is sent empty. The bandwidth is reserved strictly for that device whether used or not.

Data flow is divided into units (1 bit, 1 character, or 1 block). The multiplexer takes one unit from each line and packs them into a **frame**, which consists of one complete cycle of slots (one slot for every input line). Since we must fit n items into the time normally used to send 1, the output time slot duration becomes shorter and the unit travels faster. The frame duration is usually the same as the duration of one input unit (T).

Data Rate Relationship

$$\text{Output Data Rate} = n \times \text{Input Data Rate}$$

where n = number of input lines

For smooth flow, the speed of the main link must equal the combined speed of all inputs.

Chapter 7: Transmission Media

Transmission media are the physical pathways that carry information from source to destination. While the Physical Layer (Layer 1) controls transmission, the media itself conceptually lies below it at "Layer 0." In the broadest sense, anything that carries information qualifies as transmission media, including air for conversation or trucks for mail. In data communications, the term specifically refers to free space, metallic cable, or fiber-optic cable.

7.1 Introduction and Taxonomy

Computers transmit signals as electromagnetic energy, a combination of electric and magnetic fields vibrating in relation to one another. The electromagnetic spectrum includes power, radio waves, infrared, visible light, ultraviolet, X-rays, gamma rays, and cosmic rays. Not all portions are usable for telecommunications; we are currently limited primarily to radio frequencies and light.

Transmission media divide into two broad categories based on how the signal is constrained. **Guided media (wired)** contain the signal within a physical boundary, including twisted-pair cable, coaxial cable, and fiber-optic cable. **Unguided media (wireless)** allow the signal to travel through free space without a physical conductor, including air, vacuum, and water.

7.2 Guided Media

Guided media provide a physical conduit from one device to another, directing and containing the signal within the physical limits of the medium. Three main types exist: twisted-pair and coaxial cables (both metallic copper conductors transporting electric current) and fiber-optic cable (glass or plastic transporting light).

7.2.1 Twisted-Pair Cable

A twisted pair consists of two copper conductors, each with its own plastic insulation, twisted together. One wire carries the signal to the receiver; the second wire serves as a ground reference. The receiver uses the **difference** between the two wires to interpret data.

Why twist the wires? Signals are affected by external interference (noise) and crosstalk (signals leaking from nearby wires). If wires ran parallel, the wire closer to a noise source would pick up more interference than the farther one, creating a disparity the receiver cannot filter out. Because the wires are twisted, they swap positions regularly. Over the cable's length, both wires are equally affected by noise. Since the receiver

examines the difference between the wires and the noise is now equal on both, the noise effectively cancels itself out. The rule is simple: more twists per inch means better quality and better noise cancellation.

Unshielded Twisted-Pair (UTP) is the most common type, consisting of just twisted wires in a plastic casing. It is cheap, flexible, and easy to install, though more susceptible to noise than shielded varieties.

Shielded Twisted-Pair (STP), originally produced by IBM, encases each pair in metal foil or braided mesh that prevents noise and crosstalk penetration, but it is bulkier, heavier, and more expensive.

The EIA classifies UTP into seven categories (1 is lowest quality, 7 is highest), with each category suitable for specific uses. The most common connector is the RJ45, which features a keyed design allowing insertion in only one orientation. Performance is measured by comparing attenuation against frequency and distance. Twisted-pair cable can pass a wide range of frequencies, but attenuation increases sharply as frequency increases, especially above 100 kHz. Thicker wire gauge generally offers less resistance and better performance.

7.2.2 Coaxial Cable

Coaxial cable carries signals of higher frequency ranges than twisted-pair due to its unique layered construction. It consists of a central core (solid or stranded copper wire carrying the signal), an insulating sheath enclosing the core, an outer conductor of metal foil or braid (serving dual roles as shield against noise and second conductor completing the circuit), and a plastic outer cover.

Coaxial cables are categorized by Radio Government (RG) ratings, with each number denoting specific physical specifications. The BNC (Bayonet Neill-Concelman) connector family connects coax to devices, including the standard BNC connector for device connection, BNC T-connector for Ethernet network branching, and BNC terminator for preventing signal reflection at cable ends.

While coaxial cable offers much higher bandwidth than twisted-pair, attenuation is surprisingly higher, meaning signals weaken rapidly and require frequent repeaters. Applications include telephone networks (historically carrying up to 10,000 voice signals, now mostly replaced by fiber), cable TV (traditionally using coax throughout, now typically hybrid networks with fiber backbone and coax at the boundaries), and Ethernet LANs.

7.2.3 Fiber-Optic Cable

Fiber-optic cable, made of glass or plastic, transmits signals as light. Its operation depends on two key optical properties: **refraction** (light bending when moving from dense to less dense substances) and **reflection** (light turning back into the denser substance when hitting the boundary at a steep enough angle, called the critical angle). The fiber's core (high-density glass or plastic) is surrounded by cladding (lower density). Light traveling through the core hits the boundary and reflects back through **total internal reflection**, guiding light down the fiber like water in a pipe.

Propagation Modes

Table 7.1 Fiber-Optic Propagation Modes

Mode	Core	Light Path	Distortion	Use Case
Multimode Step-Index	Large diameter, constant density	Straight lines bouncing at edges	High distortion	Short distances
Multimode Graded-Index	Large diameter, varying density	Smooth curved paths	Less distortion	Medium distances
Single-Mode	Extremely small diameter	Nearly horizontal (one path)	Negligible	Long distances

Single-mode fiber uses a highly focused laser source. The critical angle approaches 90 degrees, so the beam travels almost horizontally with all beams arriving together, making it ideal for long-distance communication.

Fiber-optic cable composition includes the outer jacket (PVC or Teflon), Kevlar strands (the same strong material used in bulletproof vests, strengthening the cable during installation), plastic coating (cushioning the fiber), and the cladding and core (the actual optical transmission vehicle). Connectors include SC (push/pull locking, used for cable TV), ST (bayonet twist-and-lock, used for networking), and MT-RJ (same size as RJ45).

Table 7.2 Advantages and Disadvantages of Fiber-Optic

Advantages	Disadvantages
Higher bandwidth than copper	Requires specialized expertise and equipment for installation
Less signal attenuation (50 km vs 5 km without repeaters)	Unidirectional (two fibers needed for bidirectional)
Immunity to electromagnetic interference	More expensive interfaces and cable
Lightweight compared to copper	Glass splicing is harder than twisting wire
Very difficult to tap (high security)	

7.3 Unguided Media (Wireless)

Unguided media transport electromagnetic waves without a physical conductor. Signals broadcast through free space and are available to anyone with a receiving device. Signals travel in three ways depending on frequency.

Table 7.3 Signal Propagation Methods

Method	Frequency	Behavior	Range Factors
Ground Propagation	Low frequency	Waves follow earth's curvature, hugging the ground	Depends on transmission power
Sky Propagation	Higher frequency	Waves radiate to ionosphere, which reflects them back	Enables long-distance with lower power
Line-of-Sight	Very high frequency	Waves travel straight from antenna to antenna	Requires tall, facing antennas

7.3.1 Radio Waves

Radio waves operate in the 3 kHz to 1 GHz range. Their key characteristic is being **omnidirectional**: when an antenna transmits, waves travel in all directions, so sending and receiving antennas do not need to be aligned. This omnidirectional nature makes radio susceptible to interference from other antennas. Radio waves can penetrate walls (you can receive AM radio inside your house), but this penetration also means you cannot easily isolate communication to just inside a building. The available bands are relatively narrow, leading to lower digital data rates. Radio is primarily used for **multicasting** (one sender, many receivers), including AM/FM radio, television, and paging systems.

7.3.2 Microwaves

Microwaves operate in the 1 GHz to 300 GHz range. They are **unidirectional**: waves can be narrowly focused, and sending and receiving antennas must be aligned (line-of-sight). Microwaves cannot pass through walls, which makes them secure for in-building communications. The band is very wide (299 GHz), enabling high-speed transmission. However, because they travel in straight lines, the earth's curvature blocks them over long distances, requiring very tall towers and repeaters.

Microwave antennas include the **parabolic dish** (based on parabola geometry, working like a funnel to catch waves and direct them to a focus point) and the **horn antenna** (shaped like a scoop, deflecting outgoing transmissions and collecting incoming ones). Microwaves are used for **unicast** (one-to-one communication) in cellular phones, satellite networks, and wireless LANs (Wi-Fi).

7.3.3 Infrared

Infrared operates in the 300 GHz to 400 THz range and is used for short-range communication. Infrared cannot pass through walls, which prevents interference between rooms (your TV remote will not change your neighbor's channel). However, infrared cannot be used outdoors because the sun emits infrared waves that interfere with signals. The infrared band is massive (400 THz), offering potential for very high data rates.

Applications include remote controls and IrDA ports for wireless keyboards, mice, and printers (requiring line-of-sight pointing).

Table 7.4 Wireless Media Comparison Summary

Type	Frequency	Direction	Wall Penetration	Primary Usage
Radio Waves	3 kHz - 1 GHz	Omnidirectional	Yes	Multicast (Radio, TV)
Microwaves	1 GHz - 300 GHz	Unidirectional	No	Unicast (Cell phones, Wi-Fi)
Infrared	300 GHz - 400 THz	Line-of-sight	No	Short-range (Remotes)

Practice Questions & Answers

This section contains a comprehensive set of practice questions designed to test your understanding of all chapters covered in these notes. The questions range from theoretical concepts that test your understanding to numerical problems that test your ability to apply formulas. Attempt each question before checking the answer.

Theoretical Questions

Q1. What are the four fundamental characteristics of a data communication system? Explain each with an example.

Answer: The four characteristics are: (1) **Delivery** - data must reach the correct destination (e.g., an email must reach the intended recipient, not someone else); (2) **Accuracy** - data must be delivered exactly as sent (e.g., a downloaded file must match the original, bit for bit); (3) **Timeliness** - data must arrive on time (e.g., video streaming requires frames to arrive in sequence without excessive delay); (4) **Jitter** - variation in packet arrival time must be minimal (e.g., if video packets arrive at irregular intervals, playback becomes choppy and unwatchable).

Q2. List and explain the five components of a data communication system.

Answer: (1) **Message** - the information to be communicated (text, audio, video, images); (2) **Sender** - the device that sends the data (computer, phone, camera); (3) **Receiver** - the device that receives the message (computer, phone, printer); (4) **Transmission Medium** - the physical path (twisted-pair, fiber-optic, radio waves); (5) **Protocol** - the set of rules governing communication (TCP/IP, HTTP). Without a protocol, two connected devices are like two people speaking different languages - they can hear each other but cannot understand.

Q3. Compare and contrast simplex, half-duplex, and full-duplex communication modes with real-world examples.

Answer: **Simplex** is unidirectional - only one device transmits, the other only receives (like a keyboard sending to a monitor, or a radio broadcast). **Half-duplex** allows both directions but not simultaneously - devices take turns (like walkie-talkies where you say "Over" to switch). **Full-duplex** allows simultaneous two-way communication (like a telephone conversation where both parties can talk and listen at the same time).

Q4. What is a protocol? Why is it essential for communication even when two devices are physically connected?

Answer: A protocol is a set of rules that governs data communication. It is essential because physical connection alone is insufficient. A protocol defines how data is formatted, transmitted, received, and acknowledged. Without protocols, connected devices are analogous to a French speaker and a Japanese speaker in the same room - they can physically hear each other but cannot exchange meaningful information because they lack agreed-upon rules for communication.

Q5. Compare Mesh and Star topologies in terms of cost, reliability, and use cases.

Answer: **Mesh** provides a dedicated point-to-point link between every pair of devices. It is highly robust (if one link fails, alternatives exist) and private, but extremely expensive requiring $n(n-1)/2$ links for n nodes. Used in regional telephone offices. **Star** connects each device to a central hub only. It is cheaper than mesh (less cable), and a single link failure affects only that device. However, the hub represents a single point of failure. Star is commonly used in LANs.

Q6. What is the difference between a LAN and a WAN? Explain with respect to scope, connectivity, and ownership.

Answer: **Scope:** A LAN covers a single building or campus; a WAN spans towns, states, or countries. **Connectivity:** A LAN connects hosts (computers, phones) directly; a WAN connects connecting devices (routers, switches) across networks. **Ownership:** LANs are privately owned by the organization using them; WANs are typically leased from communication service providers.

Q7. Explain the difference between circuit switching and packet switching. What are the advantages and disadvantages of each?

Answer: **Circuit switching** establishes a dedicated connection reserved for the entire communication duration, like a traditional phone call. It guarantees connection quality but wastes bandwidth during silent periods. **Packet switching** breaks data into independent packets that travel separately, like postal mail. It is more efficient because lines are shared, but packets may encounter delays in queues. Circuit switching is better for real-time, constant-rate communication; packet switching is better for bursty data traffic.

Q8. What is protocol layering and why is it important? Explain with the "black box" concept.

Answer: Protocol layering divides complex communication tasks into independent layers, each with specific responsibilities. The "black box" concept means each layer is treated as a self-contained module where only inputs and outputs matter, not internal implementation. This modularity is important because changes to one layer (e.g., changing encryption methods) do not affect other layers, making the system maintainable and flexible. Without layering, a single change would require rebuilding the entire system.

Q9. State and explain the two principles of protocol layering.

Answer: Principle 1 (Bidirectional Capability): If communication is two-way, every layer must perform both operations. If Layer 3 writes, it must also read; if Layer 2 encrypts, it must also decrypt. **Principle 2 (Identical Objects):** The object exchanged between corresponding layers at sender and receiver must be identical. If Layer 2 at the sender handles ciphertext, Layer 2 at the receiver must also handle ciphertext (not plaintext or any other form).

Q10. Describe the five layers of the TCP/IP model and the function of each layer.

Answer: (1) **Physical:** Transmits individual bits across the medium, converting bits to electrical/optical signals. (2) **Data-Link:** Handles hop-to-hop delivery using frames, moving packets between adjacent nodes. (3) **Network:** Manages host-to-host communication and routing using datagrams and IP addresses. (4) **Transport:** Provides process-to-process delivery using ports; TCP offers reliability while UDP offers speed. (5) **Application:** Enables user access to network services through protocols like HTTP, SMTP, FTP, and DNS.

Q11. What is encapsulation? Describe the process as data moves from the Application layer to the Physical layer.

Answer: Encapsulation is the process of wrapping data with protocol information at each layer. At the **Application** layer, the message is created. At the **Transport** layer, a header with port numbers is added, creating a segment. At the **Network** layer, a header with IP addresses is added, creating a datagram. At the **Data-Link** layer, a header with MAC addresses and a trailer are added, creating a frame. At the **Physical** layer, the frame becomes a stream of bits. Each layer treats the data from above as payload and adds its own control information.

Q12. What is the difference between TCP and UDP? When would you use each?

Answer: **TCP** is connection-oriented, reliable, and provides flow control and error correction. It guarantees delivery but has higher overhead. Use TCP for file transfers, web browsing, and email where data integrity matters. **UDP** is connectionless, unreliable, but faster with lower overhead. Use UDP for streaming video, online gaming, and DNS queries where speed matters more than guaranteed delivery.

Q13. Compare the OSI and TCP/IP models. Why did TCP/IP succeed while OSI did not become dominant?

Answer: OSI has 7 layers (Application, Presentation, Session, Transport, Network, Data-Link, Physical) while TCP/IP has 5, combining OSI's Application, Presentation, and Session into one Application layer. TCP/IP succeeded for three reasons: (1) **Timing:** TCP/IP was already deployed when OSI arrived, making change prohibitively expensive; (2) **Incomplete specification:** OSI's Session and Presentation layers were never fully defined or implemented; (3) **Performance:** Early OSI implementations were slower than the already-working TCP/IP suite.

Q14. What is the difference between analog and digital signals? Give two examples of each.

Answer: **Analog signals** are continuous, taking infinite values within a range. Examples: human voice and temperature readings. **Digital signals** are discrete, limited to defined values (typically 0 and 1). Examples: computer memory and data stored on hard drives. An analog clock with continuously moving hands represents analog, while a digital clock with jumping numbers represents digital.

Q15. Define period, frequency, and phase. How are period and frequency related?

Answer: **Period (T)** is the time in seconds to complete one cycle. **Frequency (f)** is the number of cycles per second (Hz). **Phase** describes the waveform's position relative to time zero. Period and frequency are inverses: $f = 1/T$ and $T = 1/f$. A signal with period 0.01 s has frequency 100 Hz.

Q16. What is bandwidth? How is it calculated for both periodic and nonperiodic signals?

Answer: Bandwidth is the range of frequencies contained in a composite signal, calculated as $B = f_h - f_l$ where f_h is the highest and f_l is the lowest frequency. For periodic signals, bandwidth includes discrete frequency spikes (harmonics). For nonperiodic signals, bandwidth encompasses a continuous range of frequencies.

Q17. Explain the difference between time domain and frequency domain representations of a signal.

Answer: The **time domain** plots amplitude versus time, showing how a signal changes moment by moment. It is intuitive but makes it hard to read frequency. The **frequency domain** plots peak amplitude versus frequency, showing the signal's component frequencies as spikes. It is compact and makes analyzing complex signals easier. A sine wave in time domain is a curve; in frequency domain it is a single spike.

Q18. What is a composite signal? Explain Fourier analysis and its significance in data communications.

Answer: A composite signal is made of multiple simple sine waves combined. Jean-Baptiste Fourier proved that *any* composite signal can be decomposed into simple sine waves with different frequencies, amplitudes, and phases. This is significant because it means any complex signal (including digital signals) can be analyzed as a sum of simple sine waves, which is fundamental to understanding bandwidth requirements and signal processing.

Q19. What is the difference between baseband and broadband transmission? When is each used?

Answer: **Baseband** sends digital signals without converting to analog, requiring a low-pass channel (bandwidth starts at 0). Used in LANs with dedicated cables. **Broadband** converts digital to analog through modulation, allowing use of bandpass channels (bandwidth does not start at 0). Used when low-pass channels are unavailable, such as over telephone lines (modems) or cellular networks.

Q20. Define attenuation, distortion, and noise. List the four types of noise.

Answer: **Attenuation** is loss of signal energy due to medium resistance (fixed with amplifiers). **Distortion** is change in signal shape due to different frequency components arriving at different times. **Noise** is unwanted external energy corrupting the signal. The four noise types are: (1) Thermal noise (electron motion in wires), (2) Induced noise (from motors/appliances), (3) Crosstalk (interference between adjacent wires), (4) Impulse noise (sudden spikes from lightning/power surges).

Q21. What is SNR? Explain the difference between SNR and SNR in decibels.

Answer: SNR (Signal-to-Noise Ratio) is the ratio of average signal power to average noise power ($\text{SNR} = P_{\text{signal}} / P_{\text{noise}}$). SNR in decibels is $\text{SNR}_{dB} = 10 \log_{10}(\text{SNR})$. If $\text{SNR} = 10,000$, then $\text{SNR}_{dB} = 40 \text{ dB}$. The decibel version compresses large ratios into manageable numbers. For Shannon capacity calculations, you must convert dB back to the raw ratio first.

Q22. State the Nyquist Bit Rate formula and the Shannon Capacity formula. What is the key difference between them?

Answer: **Nyquist:** $\text{BitRate} = 2 \times B \times \log_2(L)$ - determines maximum bit rate for a noiseless channel based on bandwidth and signal levels. **Shannon:** $C = B \times \log_2(1 + \text{SNR})$ - determines absolute maximum capacity for a noisy channel. The key difference: Nyquist includes signal levels (L) and assumes no noise; Shannon accounts for noise (via SNR) but does not include signal levels. Nyquist tells us what encoding to use; Shannon tells us the channel's upper limit.

Q23. What is multiplexing? Why is it necessary? Name the three basic techniques.

Answer: Multiplexing allows simultaneous transmission of multiple signals across a single data link. It is necessary because bandwidth is limited and expensive; adding physical cables for every new channel is impractical. The three techniques are: **FDM (Frequency-Division Multiplexing)** for analog signals (shares frequency), **WDM (Wavelength-Division Multiplexing)** for optical signals (shares light wavelengths), and **TDM (Time-Division Multiplexing)** for digital signals (shares time slots).

Q24. Explain FDM and TDM with analogies. How do they differ in their approach to sharing a link?

Answer: **FDM** is like FM radio stations: all stations broadcast simultaneously through the air on different frequencies without interfering. It divides bandwidth horizontally into frequency bands. **TDM** is like a single-lane road where cars take turns: each user gets the entire bandwidth for a short time slot. It divides the link vertically into time slots. FDM is used for analog signals; TDM for digital signals.

Q25. What is WDM and how does it differ from FDM?

Answer: WDM (Wavelength-Division Multiplexing) is conceptually identical to FDM but operates at much higher frequencies (light waves in fiber-optic cables). While FDM combines radio frequencies, WDM combines different wavelengths (colors) of light using prisms. WDM is specifically designed for fiber-optic cables and can carry much more data than FDM due to the enormous bandwidth of light frequencies.

Q26. Compare twisted-pair, coaxial, and fiber-optic cables in terms of bandwidth, attenuation, cost, and immunity to noise.

Answer: **Twisted-pair:** Lowest bandwidth, moderate attenuation, cheapest, susceptible to noise (improved by twisting and shielding). **Coaxial:** Higher bandwidth than twisted-pair, surprisingly high attenuation, moderate cost, better noise immunity due to shielding. **Fiber-optic:** Highest bandwidth, very low attenuation (can go 50 km without repeaters), most expensive, complete immunity to electromagnetic interference.

Q27. Why are the wires in a twisted-pair cable twisted? Explain the noise cancellation mechanism.

Answer: Twisting causes both wires to be equally exposed to external noise over the cable length. In one twist, wire A is closer to the noise source; in the next twist, wire B is closer. Since the receiver interprets the *difference* between the two wires, and noise affects both equally, the noise cancels out. More twists per inch provides better noise cancellation. Without twisting, parallel wires would pick up unequal noise, creating a differential that corrupts the signal.

Q28. Explain how fiber-optic cable uses total internal reflection to guide light signals.

Answer: Fiber-optic cable has a high-density core surrounded by lower-density cladding. When light traveling through the core hits the core-cladding boundary at an angle greater than the critical angle, it reflects back into the core rather than passing through. This **total internal reflection** traps light inside the core, guiding it down the fiber like water in a pipe. In single-mode fiber, the small core diameter ensures the light travels almost straight with minimal bouncing, reducing distortion.

Q29. Compare radio waves, microwaves, and infrared in terms of frequency range, directionality, and applications.

Answer: **Radio waves** (3 kHz - 1 GHz) are omnidirectional, penetrate walls, used for multicasting (AM/FM radio, TV). **Microwaves** (1 GHz - 300 GHz) are unidirectional, require line-of-sight, cannot penetrate walls, used for unicasting (cell phones, Wi-Fi, satellite). **Infrared** (300 GHz - 400 THz) is short-range line-of-sight, cannot penetrate walls, used for remotes and IrDA ports.

Q30. What is the difference between a link and a channel in the context of multiplexing?

Answer: A **link** is the physical pathway (the actual cable or medium). A **channel** is the portion of a link that carries a specific transmission. One link can carry multiple channels simultaneously through multiplexing. For example, a single fiber-optic link might carry 100 channels using WDM, where each channel is a different wavelength of light.

Numerical Problems

N1. A signal has a period of 20 ms. What is its frequency in Hz and kHz?

Solution:

$$T = 20 \text{ ms} = 20 \times 10^{-3} \text{ s} = 0.02 \text{ s}$$

$$f = 1/T = 1/0.02 = 50 \text{ Hz} = 50 \times 10^{-3} \text{ kHz} = 0.05 \text{ kHz}$$

Answer: 50 Hz or 0.05 kHz

N2. A signal has a frequency of 500 kHz. What is its period in microseconds?

Solution:

$$f = 500 \text{ kHz} = 500 \times 10^3 \text{ Hz} = 5 \times 10^5 \text{ Hz}$$

$$T = 1/f = 1/(5 \times 10^5) = 2 \times 10^{-6} \text{ s} = 2 \text{ us}$$

Answer: 2 microseconds

N3. Express a period of 50 ms in microseconds and in seconds.

Solution:

$$50 \text{ ms} = 50 \times 1000 \text{ us} = 50,000 \text{ us}$$

$$50 \text{ ms} = 50 \times 10^{-3} \text{ s} = 0.05 \text{ s}$$

Answer: 50,000 microseconds or 0.05 seconds

N4. If the RMS voltage in a US home is 120 V, what is the peak amplitude?

Solution:

$$\text{Peak} = \sqrt{2} \times \text{RMS}$$

$$\text{Peak} = \sqrt{2} \times 120 = 1.414 \times 120 = 169.7 \text{ V}$$

Answer: Approximately 169.7 V (or about 170 V)

N5. A sine wave is offset by 1/4 cycle with respect to time zero. What is its phase shift?

Solution:

$$\text{Phase} = (1/4) \times 360 \text{ degrees} = 90 \text{ degrees}$$

Answer: 90 degrees

N6. A periodic signal has sine waves at frequencies 200 Hz, 600 Hz, and 1000 Hz. What is its bandwidth?

Solution:

$$f_l = 200 \text{ Hz}, f_h = 1000 \text{ Hz}$$

$$B = f_h - f_l = 1000 - 200 = 800 \text{ Hz}$$

Answer: 800 Hz

N7. A signal has a bandwidth of 50 Hz. The highest frequency is 180 Hz. What is the lowest frequency?

Solution:

$$\begin{aligned} B &= f_h - f_l \\ 50 &= 180 - f_l \\ f_l &= 180 - 50 = 130 \text{ Hz} \end{aligned}$$

Answer: 130 Hz

N8. A periodic composite signal has a bandwidth of 1500 Hz, centered at 2500 Hz. What are the lowest and highest frequencies?

Solution:

$$\begin{aligned} B &= 1500 \text{ Hz, middle frequency} = 2500 \text{ Hz} \\ B \text{ extends } B/2 &= 750 \text{ Hz above and below} \\ f_l &= 2500 - 750 = 1750 \text{ Hz} \\ f_h &= 2500 + 750 = 3250 \text{ Hz} \end{aligned}$$

Answer: Lowest = 1750 Hz, Highest = 3250 Hz

N9. If a signal has 8 levels, how many bits can be sent per level?

Solution:

$$\text{Bits per level} = \log_2(L) = \log_2(8) = 3 \text{ bits}$$

Answer: 3 bits per level

N10. How many signal levels are needed to send 4 bits per level?

Solution:

$$\begin{aligned} L &= 2^n \text{ where } n = \text{bits per level} \\ L &= 2^4 = 16 \text{ levels} \end{aligned}$$

Answer: 16 levels

N11. A digital signal has a bit rate of 2000 bps. If the propagation speed is 2×10^8 m/s, what is the bit length?

Solution:

```
Bit duration = 1 / bit rate = 1 / 2000 = 0.0005 s =  $5 \times 10^{-4}$  s
Bit length = Propagation speed x Bit duration
Bit length =  $2 \times 10^8 \times 5 \times 10^{-4} = 10^5$  m = 100 km
```

Answer: 100 km (or 100,000 meters)

N12. We need to send 200 kbps over a low-pass channel using baseband transmission. What is the minimum bandwidth required (rough approximation)? What bandwidth is needed for better accuracy (using 1st and 3rd harmonics)?

Solution:

```
Rough (minimum):  $B = N/2 = 200/2 = 100$  kHz
Better (1st + 3rd harmonics):  $B = 3N/2 = 3 \times 100 = 300$  kHz
```

Answer: Minimum = 100 kHz; Better = 300 kHz

N13. A signal loses 3 dB of power after passing through a medium. If the input power is 10 mW, what is the output power?

Solution:

```
dB =  $10 \log_{10}(P2/P1)$ 
-3 =  $10 \log_{10}(P2/10)$ 
-0.3 =  $\log_{10}(P2/10)$ 
 $10^{-0.3} = P2/10$ 
0.501 =  $P2/10$ 
 $P2 = 5.01$  mW
```

Answer: Approximately 5 mW (the signal lost about half its power)

N14. A signal passes through three amplifiers with gains of +5 dB, -2 dB, and +8 dB. What is the total gain in dB? Has the signal been amplified or attenuated overall?

Solution:

$$\text{Total dB} = 5 + (-2) + 8 = 11 \text{ dB}$$

Answer: +11 dB total. Since the result is positive, the signal has been **amplified** overall.

N15. A channel has a bandwidth of 2 kHz and an SNR of 30 dB. What is the Shannon capacity?

Solution:

Step 1: Convert SNR from dB to ratio:

$$30 = 10 \log_{10}(\text{SNR})$$

$$\text{SNR} = 10^3 = 1000$$

Step 2: Apply Shannon's formula:

$$C = B \times \log_2(1 + \text{SNR})$$

$$C = 2000 \times \log_2(1 + 1000)$$

$$C = 2000 \times \log_2(1001)$$

$$C = 2000 \times 9.967$$

$$C = 19,934 \text{ bps} = 19.93 \text{ kbps}$$

Answer: Approximately 19.93 kbps (approximately 20 kbps)

N16. A noiseless channel has a bandwidth of 5 kHz. If we use 8 signal levels, what is the maximum bit rate according to Nyquist?

Solution:

$$\text{BitRate} = 2 \times B \times \log_2(L)$$

$$\text{BitRate} = 2 \times 5000 \times \log_2(8)$$

$$\text{BitRate} = 2 \times 5000 \times 3$$

$$\text{BitRate} = 30,000 \text{ bps} = 30 \text{ kbps}$$

Answer: 30 kbps

N17. A channel has a bandwidth of 1 MHz and SNR = 40 dB. (a) Find the Shannon capacity. (b) If we want to achieve this capacity, how many signal levels are needed according to Nyquist?

Solution:

(a) Convert SNR: $40 = 10 \log_{10}(\text{SNR}) \Rightarrow \text{SNR} = 10^4 = 10,000$
 $C = 10^6 \times \log_2(1 + 10000)$
 $C = 10^6 \times \log_2(10001)$
 $C = 10^6 \times 13.29 = 13.29 \text{ Mbps}$

(b) Using Nyquist: $13.29 \times 10^6 = 2 \times 10^6 \times \log_2(L)$
 $\log_2(L) = 6.64$
 $L = 2^{6.64} = 99.4 \approx 100 \text{ levels}$

Answer: (a) 13.29 Mbps; (b) Approximately 100 signal levels

N18. We need to upgrade the bit rate of a channel from 1 Mbps to 4 Mbps. If the original signal used 4 levels, how many levels are needed for the upgraded rate (assuming same bandwidth and noiseless channel)?

Solution:

Original: $1 \times 10^6 = 2 \times B \times \log_2(4) = 2 \times B \times 2$
 So $B = 250 \text{ kHz}$

 New: $4 \times 10^6 = 2 \times 250,000 \times \log_2(L)$
 $4 \times 10^6 = 500,000 \times \log_2(L)$
 $\log_2(L) = 8$
 $L = 2^8 = 256 \text{ levels}$

Answer: 256 levels

N19. Five channels, each with a bandwidth of 200 kHz, are to be multiplexed using FDM. If we need 20 kHz guard bands between channels, what is the minimum bandwidth required for the multiplexed link?

Solution:

```
Total signal bandwidth = 5 x 200 kHz = 1000 kHz
Number of guard bands needed = 5 - 1 = 4
Total guard band bandwidth = 4 x 20 kHz = 80 kHz
Minimum link bandwidth = 1000 + 80 = 1080 kHz = 1.08 MHz
```

Answer: 1.08 MHz

N20. In a synchronous TDM system, 4 input lines each send data at 9600 bps. What is the minimum data rate of the multiplexed output line?

Solution:

```
Output Data Rate = n x Input Data Rate
Output Data Rate = 4 x 9600 = 38,400 bps = 38.4 kbps
```

Answer: 38.4 kbps

N21. A mesh network has 8 devices. How many duplex links are needed?

Solution:

```
Number of links = n(n-1)/2
Links = 8 x 7 / 2 = 56 / 2 = 28 links
```

Answer: 28 duplex links

N22. A fiber-optic cable needs to send data at 1 Gbps. If light travels at 2×10^8 m/s in the fiber, what is the bit length?

Solution:

```
Bit duration = 1 / (1 x 10^9) = 10^-9 s = 1 ns
Bit length = 2 x 10^8 x 10^-9 = 0.2 m = 20 cm
```

Answer: 20 cm per bit

N23. An AM radio station operates at 1000 kHz. If each station is allocated 10 kHz, what is the frequency range of this station's broadcast?

Solution:

Center frequency = 1000 kHz
Bandwidth per station = 10 kHz
Range: $1000 - 5 = 995$ kHz to $1000 + 5 = 1005$ kHz

Answer: 995 kHz to 1005 kHz

N24. A signal travels through a medium and its power drops to one-tenth of its original value. Express this attenuation in decibels.

Solution:

$P_2/P_1 = 1/10 = 0.1$
 $\text{dB} = 10 \log_{10}(0.1) = 10 \times (-1) = -10$ dB

Answer: -10 dB (10 dB of attenuation/loss)

N25. A periodic signal completes 500 cycles in 2 milliseconds. What is its frequency?

Solution:

Frequency = cycles / time = $500 / (2 \times 10^{-3})$
 $f = 500 / 0.002 = 250,000$ Hz = 250 kHz

Answer: 250 kHz